Ogbonna Arinze

arinzeogbo@gmail.com | LinkedIn

SUMMARY

Web Developer with 8 years of professional experience building modern, high-performance web applications using React, Next.js, and TypeScript across diverse industries including healthcare, fintech, sports, and civic tech. Proven expertise in full-stack architecture with Next.js App Router, Server-Side Rendering, and scalable component systems, paired with deep CI/CD, automated testing, and cloud deployment experience on AWS and Vercel. Passionate about delivering polished digital products, optimizing performance, and collaborating cross-functionally with design and backend teams in fast-paced, agency-style environments.

CORE COMPETENCIES

- **Frontend Development:** React, Next.js (App Router, SSR, SSG), TypeScript, JavaScript (ESNext), Vue.js, Tailwind CSS, and Radix UI.
- **Full-Stack Architecture:** NestJS, NodeJS REST APIs, GraphQL, PostgreSQL, MongoDB, Prisma, JWT authentication, and RBAC.
- **Testing & Quality:** Playwright (E2E), Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA pipelines.
- **Cloud & CI/CD:** AWS (CloudFront, S3), Vercel, Docker, GitHub Actions, Nx monorepos, pnpm workspaces, and Infrastructure as Code.
- **Performance & Accessibility:** Core Web Vitals, Lighthouse, code splitting, lazy loading, CDN strategies, bundle analysis, and WCAG compliance.
- **AI & Innovation:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, vector search (pgvector, Chroma), and agentic AI workflows.
- **Leadership:** Technical architecture reviews, mentorship, Agile/Scrum, cross-functional collaboration, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query them in real time via a conversational interface.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations and enhancing adoption across teams.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Implemented **JWT-based authentication** and **role-based access control** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **GitHub Actions CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality through structured code reviews.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports and more reliable release cycles.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices across the team.

- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**, using **React**, **Redux**, **RxJS**, and **Axios**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) with complex state management using **Redux Toolkit** and real-time data streaming via **RxJS**.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Ant Design** UI components for role and permission management.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and structured task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, and converting business requirements into database schemas and comprehensive application documentation.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with **React 19** and **App Router**, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm workspaces**.
- Built the **Next.js** citizen-facing frontend with **Tailwind CSS v4** and **Recharts**, featuring 22 interactive chart types auto-generated from AI responses, dark mode, URL state management with nuqs, **Server-Side Rendering** for public pages, and responsive mobile layouts.
- Designed and implemented a **multi-agent AI system** using the Mastra framework with 5 specialized agents and a three-stage intent classification pipeline, powered by a **hybrid RAG pipeline** combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and **Traefik** reverse proxy routing.
- Implemented **SSE streaming** for real-time chat, **PostgreSQL 16** with **Prisma v7**, phone OTP and Telegram OAuth authentication, and shareable public conversations with **SEO optimization** (Open Graph metadata, JSON-LD schema, slug-based URLs with SSR).

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*